Artificial Intelligence (61FIT3AIN), midterm project 12

Campus FAQ Assistant

Finding the right answer among 112 campus FAQs and notices with TF-IDF, Logistic Regression and explicit rules for saying “not found”.

How do I reset my Sutdent Portal password?

faq_technical_support_01
How do I reset my Student Portal password?

cosine 0.53
technical_support, 90% confidence

Nguyễn Đạt Vinh (team leader) 2301040198
Lương Việt Anh 2301040004
Nguyễn Ánh Dương 2401040080

Hanoi University, November 2026

The system must find the right document, or say plainly that none exists

What the user types	The system must	Expected result
`How do I reset my Student Portal password?`	Find the right FAQ	faq_technical_support_01
`How do I reset my Sutdent Portal password?`	Still find it despite the typo	faq_technical_support_01
`What's the best restaurant near campus for vegan food?`	Abstain, never make one up	Information not found

`classify_intent(query)`

Returns one of 7 intents with a confidence score, e.g. `("exam", 0.82)`.

`retrieve(query, top_k)`

Returns up to k (document, score) pairs, or an empty list when nothing is relevant enough.

Project constraints

Classical methods only, run offline with scikit-learn, no LLM APIs.

The 120 test queries target three weak spots of word matching

The provided data

112 documents: 84 FAQs and 28 notices, split evenly across 7 intents (16 each): course_registration, exam, tuition, class_schedule, student_services, graduation, technical_support.

The 120 test queries



35	35	35	15
normal	paraphrase	typo	out of corpus

normal = verbatim FAQ question; each normal query also has one paraphrase and one typo version.

Vocabulary mismatch (paraphrase)

Same meaning, different words, so word matching sees no overlap.

`section is full` vs `no open seats`

Typos

Matched as whole words, “conncet” is an entirely new word.

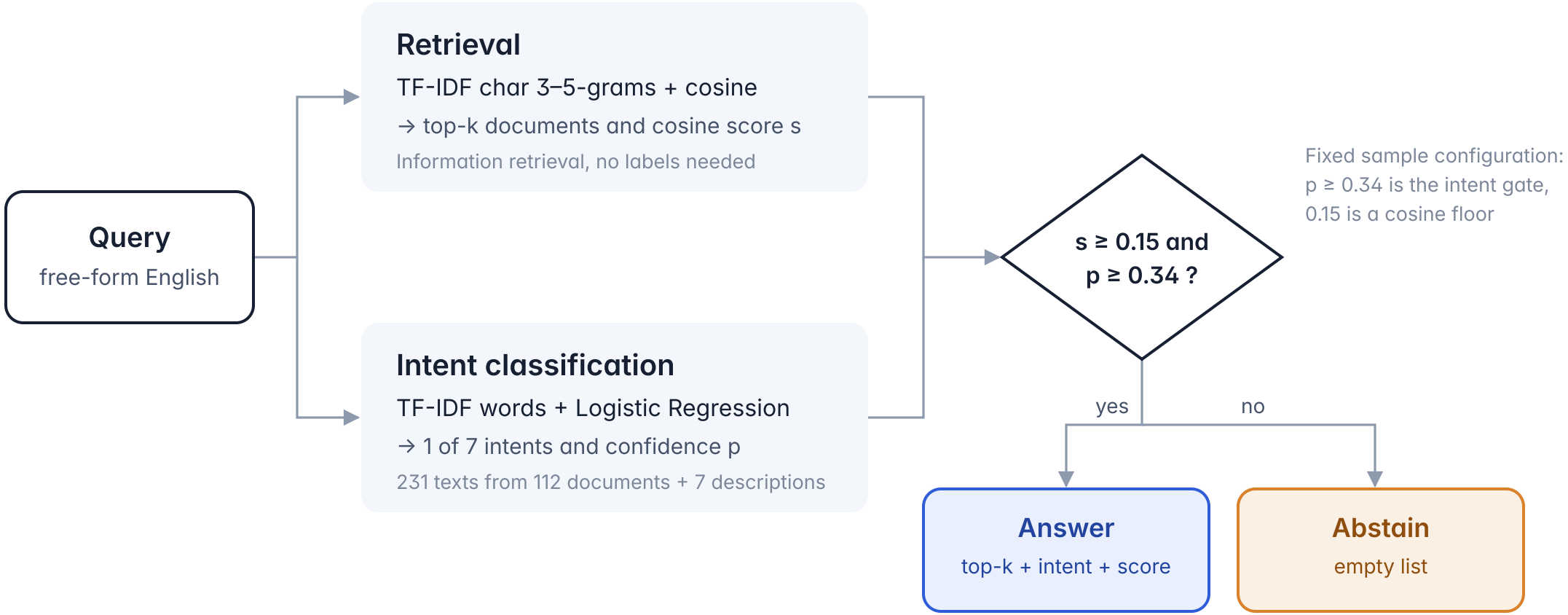
`connect` vs `conncet`

Accidental overlap (out of corpus)

An unrelated question may share features with the corpus and receive a nonzero score.

`Can I rent a kayak from the campus recreation center?`

Retrieval and intent provide two scores for the abstention decision



Train: 112 text + 112 question/title + 7 descriptions. p = max posterior over the 7 intents.

TF-IDF down-weights common words; cosine measures direction, not length

$$\text{tfidf}(t,d) = \text{tf}(t,d) \times \left(\ln \frac{1+N}{1+\text{df}(t)} + 1 \right)$$

Sublinear TF = 1 + ln(count) for count > 0. IDF uses smoothing; vectors are L2-normalized. N = 112.

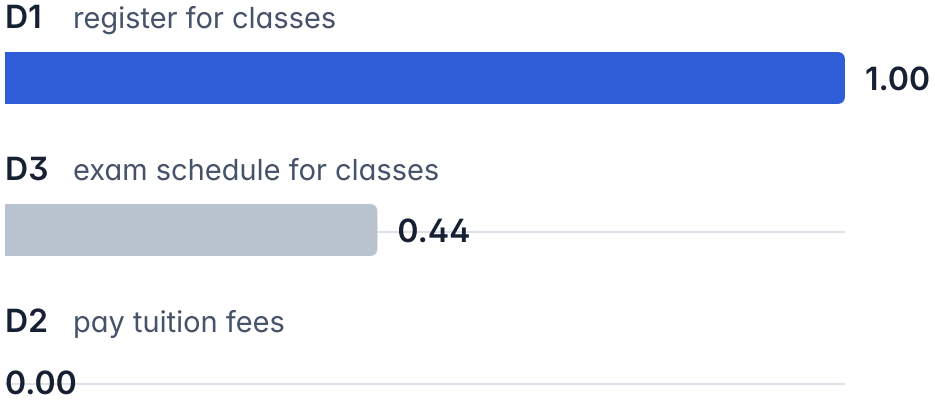
$$\cos\left(\vec{q},\vec{d}\right) = \frac{\vec{q}\cdot\vec{d}}{\|\vec{q}\|\ \|\vec{d}\|}$$

Cosine compares direction after length normalization. Adding new features can still change the score.

> Character 3–5-grams: “connect” and “conncet” share no word, but still share “ co”, “con”, “onn”.

Worked example with the same smoothed IDF, N = 3

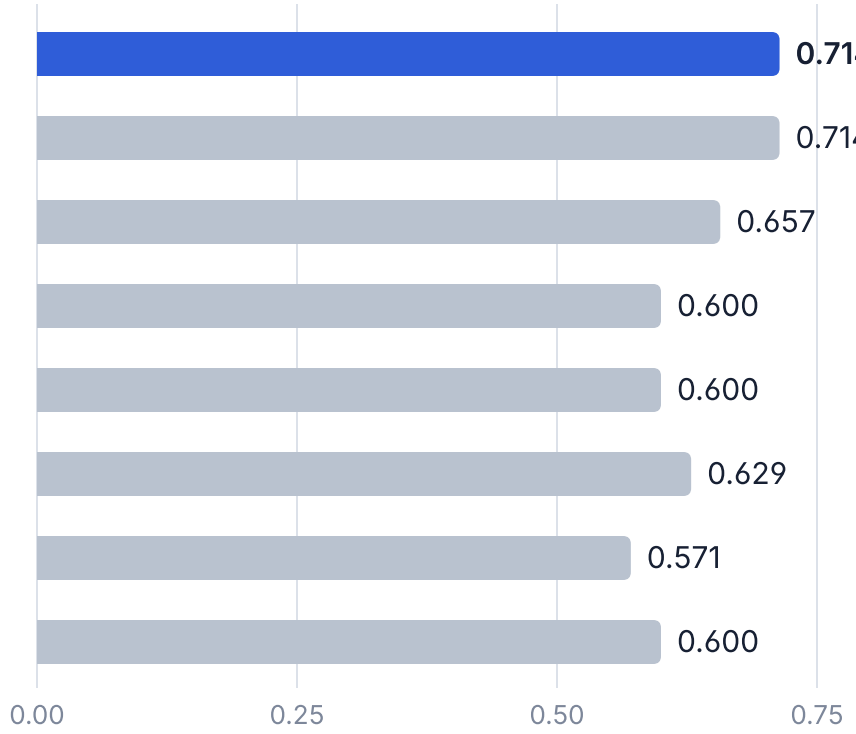
Query `how to register for classes`



> Using the smoothed IDF formula on the left, D3 scores 0.44 and ranks below D1. The query is projected onto the corpus vocabulary.

Char 3–5 TF-IDF leads the eight configurations shown, with MRR 0.943

Method (sorted by MRR)	P@1 on 35 paraphrased queries	MRR, 105 queries
TF-IDF char 3–5-grams	**0.714**	**0.943**
Hybrid: 0.5 word + 0.5 char	0.714	0.936
TF-IDF words	0.657	0.927
BM25 (k1 = 1.5, b = 0.75)	0.600	0.912
TF-IDF words, 1–2-grams	0.600	0.905
TF-IDF words, no stopwords + sublinear	0.629	0.896
Bag-of-Words + cosine (baseline)	0.571	0.893
LSA (TF-IDF + 60-dim SVD)	0.600	0.873

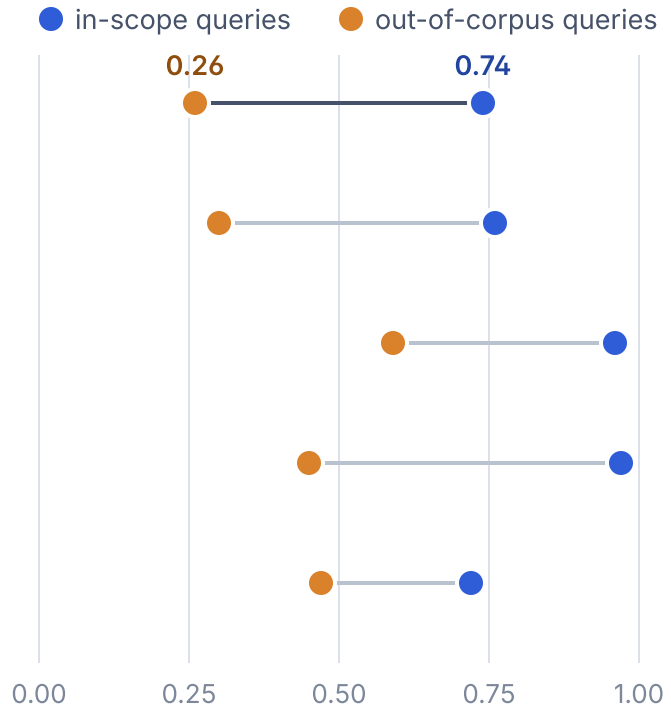


> Char 2–4 scored MRR 0.953 and P@1 92.4% on the same benchmark. We retain 3–5 for consistency with the sample code and demo; this is not an independent model-selection result.

Comparison table from the team's research document; the final configuration is reproduced by scripts/evaluate.py.

Logistic Regression gets 97.1% of intents right and is least confident on out-of-corpus queries

Model	Accuracy	Mean confidence
TF-IDF + Logistic Regression	**97.1%**	0.26 → 0.74
TF-IDF char + Logistic Regression	97.1%	
TF-IDF char + Naive Bayes	96.2%	
BoW + Naive Bayes (baseline)	94.3%	
kNN (k = 5, cosine)	87.6%	



$$P\left(c \mid \vec{x}\right) = \frac{\exp\left(\vec{w}_c \cdot \vec{x} + b_c\right)}{\sum_k \exp\left(\vec{w}_k \cdot \vec{x} + b_k\right)}$$

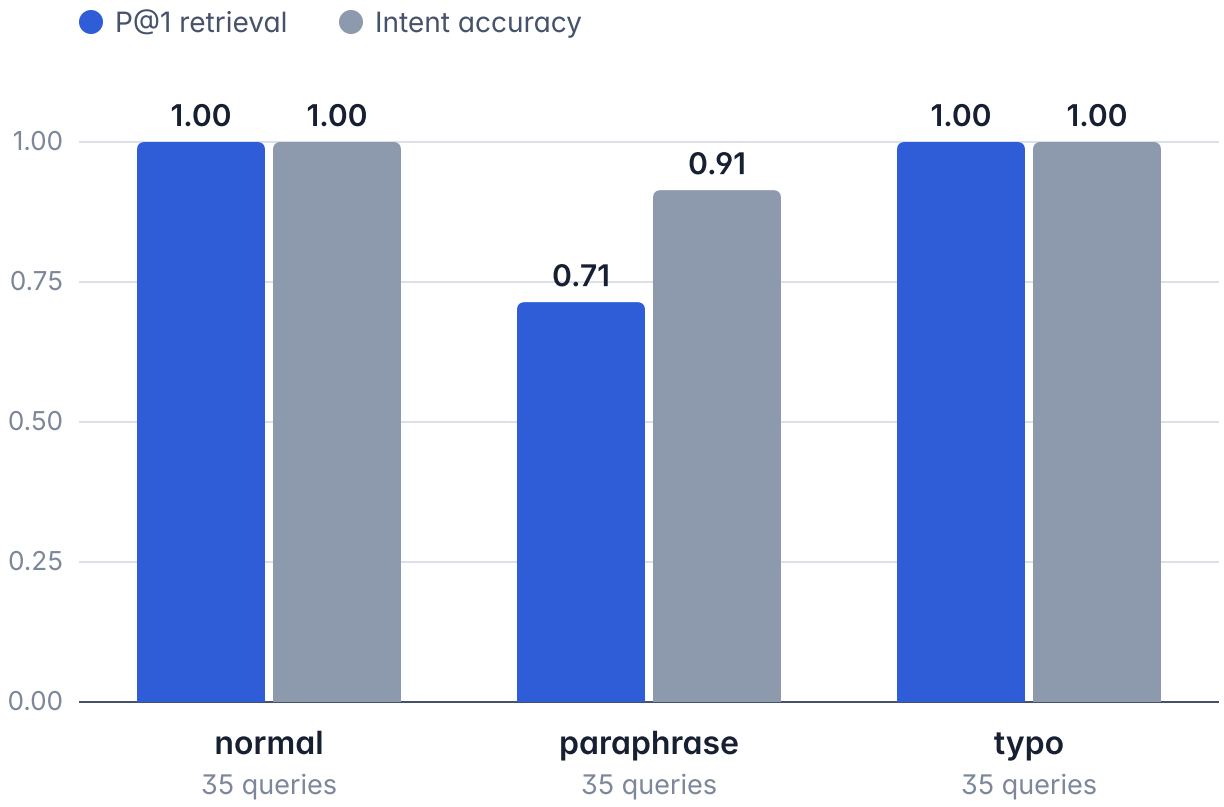
Logistic Regression is a **discriminative** model: it learns the boundaries between the 7 intents directly.

Naive Bayes is **generative** and assumes conditional independence. BoW + NB gives mean OOC confidence 0.45 on this benchmark.

LinearSVC provides decision scores; probabilities require calibration. LR confidence helps reject these OOC queries, but its calibration has not been measured.

Confidence is a model posterior over 7 intents. The chart does not establish probability calibration.

Typos barely matter; paraphrases cut P@1 to 0.71



Why typos are mild

The project's generator only swaps two adjacent letters in one word. The correct document's cosine drops by about 0.05, but its rank stays the same.

Why paraphrases are hard

Same meaning, few shared words. Still, 33 of 35 have the correct document in the top 3.

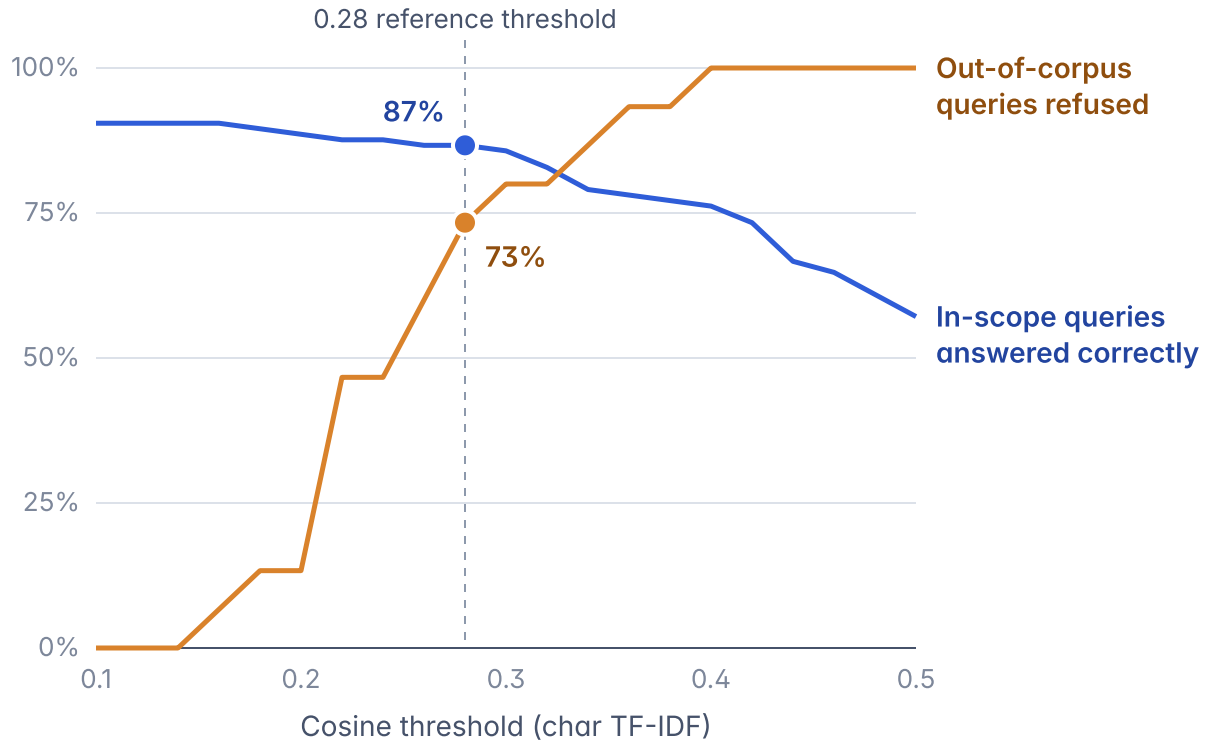
Four metrics by query type

Query type	MRR	P@1	P@3	Intent
normal	1.000	1.000	0.333	100%
paraphrase	0.829	0.714	0.314	91.4%
typo	1.000	1.000	0.333	100%

P@3 is at most 0.333 because each query has exactly one correct document.

All 35 normal queries occur in classifier training. Paraphrase intent accuracy: 32/35. Fixed-configuration benchmark.

The intent gate gives the observed improvement; cosine remains a similarity floor



Gate ablation with fixed thresholds

Abstention rule	Correct / 120	OOC
No abstention	95	0/15
Cosine only, s ≥ 0.15	96	1/15
Intent only, p ≥ 0.34	108	15/15
Both gates	108	15/15

> The intent gate gives the observed gain. Adding the cosine floor does not change accuracy here. At cosine 0.28 alone: 102/120 correct, 11/15 OOC rejected.

Reproduced from the integrated code on the reused benchmark. Fixed configuration; not a cross-validation estimate.

All 10 ranking errors are paraphrases: the model matches words, not meaning

`Two of my finals overlap on the same day - who do I tell?`

Expected faq_exam_02: two of my final exams are scheduled at the same time

Returned faq_class_schedule_02: two of my classes overlap in time

`What's the process for signing up for courses at Hanoi University?`

Expected faq_course_registration_01: how do I register for classes

Returned faq_class_schedule_08: class schedule ... university holidays

`My laptop won't join the campus Wi-Fi - what should I check?`

Expected faq_technical_support_02: can't connect to the campus Wi-Fi

Returned notice_technical_support_04: Laptop Loan Program

7 of 10 errors are off by one rank

The correct document is ranked 2nd, so showing the top 3 already surfaces the answer.

Fixes for the final version

A domain synonym list (full ↔ no open seats, conflict ↔ overlap), more weight on the question than on the answer text, and searching only within the predicted intent.

Live demo: 3 correct answers, 2 correct abstentions, 1 deliberate failure

Typed query	Result	What to point out
`How do I reset my Student Portal password?`	faq_technical_support_01, 0.54	Standard case
`How do I reset my Sutdent Portal password?`	faq_technical_support_01, 0.53	The typo only lowers the score slightly
`I forgot my Student Portal password - how do I reset it?`	faq_technical_support_01, 0.51	Reworded, but enough shared words
`What's the best restaurant near campus for vegan food?`	Abstain, 31% confidence	The intent gate blocks it
`Can I rent a kayak from the campus recreation center?`	Abstain, 29% confidence	Cosine 0.39; only the intent gate catches it
`Two of my finals overlap on the same day - who do I tell?`	Wrong: faq_class_schedule_02	Doesn't know overlap = conflict

Our three final choices, and what the numbers don't show

Step	Choice	Result
Retrieval	TF-IDF char 3–5-grams + cosine	MRR 0.943
Intent	TF-IDF + Logistic Regression	97.1%
Abstention	Cosine floor + intent confidence	90% (108/120)

The selected sample is lightweight, explainable and within the classical-method constraints. The intent gate gives the observed gain; the cosine gate supplies a similarity floor.

Next steps for the final

- A domain synonym list to handle paraphrases.
- Separate weights for the question and the answer text.
- A Streamlit interface with a threshold slider and source citations.

Limitations

- The sample configuration was developed with this benchmark in view. Generalization to fresh queries has not been measured.
- Only 15 out-of-corpus queries; the most confident one scores 0.33, just under the 0.34 threshold.
- The test typos are mild; heavier typos were not tested.
- The model does not understand synonyms: only 71% of paraphrases are right at rank 1.

Thank you

Questions for our team...

Nguyễn Đạt Vinh, **Lương Việt Anh**, **Nguyễn Ánh Dương**

Code and data: starter/student_core.py, scripts/evaluate.py